

NHS Expenditure Analysis: Comprehensive Documentation

1. Project Overview

This analysis explores NHS expenditure trends over time using three key datasets:

- **NHS Expenditure by Budget Category and Year** (Total Expenditure)
- **NHS Expenditure Per Head by Budget Category and Year** (Per Capita Spending)
- **NHS Expenditure as a Percentage of Total by Budget Category and Year** (Relative Budget Allocation)

The objective is to assess NHS spending patterns, efficiency, and priority shifts, particularly around the COVID-19 period.

2. Key Analytical Steps

2.1 Data Preprocessing

- **Dropped unnecessary columns** (LHB primary/secondary and other breakdowns).
- **Filtered relevant data** (keeping only rows with "Totals" in Broad Category).
- **Validated data integrity** (checked missing values, column structures).

2.2 Total NHS Expenditure Analysis

- **Trend Analysis:** NHS spending increased steadily from 2009-10, with a sharp rise in 2020-21 (COVID-19 impact).
- **Category-Wise Analysis:** Largest spending categories include **Mental Health, Cancers & Tumours, and Infectious Diseases**.
- **Statistical Tests:**
 - Welch's t-Test confirmed **significant expenditure shifts** pre- and post-pandemic.
 - Levene's Test detected **increased volatility** in spending trends.

2.3 Per Capita NHS Spending Analysis

- **Trend Analysis:** Per capita spending followed total NHS expenditure closely, suggesting efficient scaling.
- **Growth Rate Comparison:**
 - A spike in **2020-21**, but remains higher than pre-pandemic levels.
- **Consistent alignment between total and per capita growth suggests controlled financial planning.**
- **Statistical Tests:**
 - Welch's t-Test confirmed a **statistically significant increase in per capita expenditure**.
 - Variance analysis suggested spending volatility remained **consistent post-pandemic**.

2.4 NHS Budget Allocation as a Percentage of Total

- **Major Budget Gainers and Losers:**

- **Biggest Increase:** Infectious Diseases (COVID-19 funding surge).
 - **Biggest Decrease:** Musculo Skeletal System Problems (possible deprioritization).
 - **Heatmap Analysis:** Showed a clear funding reallocation in 2020-21.
 - **Statistical Tests:**
 - Welch's t-Test confirmed **statistically significant shifts** in budget allocation for Musculo Skeletal System Problems.
 - **Variance remained stable**, indicating planned policy changes rather than reactive shifts.
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3. Visualizations

To effectively present findings, we incorporated diverse visualization techniques:

- **Line Charts:** Trend analysis for total, per capita, and category-wise expenditure.
 - **Heatmaps:** Highlighted shifts in NHS priority over time.
 - **Stacked Area Chart:** Provided an overview of how different categories evolved relative to total spending.
 - **Bar Charts:** Compared pre- and post-pandemic spending by category.
 - **Boxplots:** Assessed spending volatility across NHS budget categories.
 - **Correlation Heatmaps:** Identified relationships between different NHS spending categories.
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4. Key Findings & Implications

4.1 NHS Spending Trends

- **Total NHS spending and per capita expenditure increased consistently.**
- **Sharpest growth occurred in 2020-21, aligning with pandemic response measures.**
- **Spending growth remained high post-pandemic, suggesting lasting structural changes.**

4.2 Budget Reallocation Trends

- **Infectious Diseases funding saw the largest increase, but stabilized post-pandemic.**
- **Musculo Skeletal System Problems experienced a statistically significant decline.**
- **NHS spending volatility did not increase post-pandemic, suggesting planned reallocations rather than reactive fluctuations.**

4.3 Policy & Efficiency Insights

- **Efficiency Findings:**
 - Pre- and post-pandemic NHS spending trends suggest an efficient scaling of resources.
 - Growth rates of total and per capita spending remained aligned, reducing risk of inefficient allocation.
 - **Long-Term Strategic Shifts:**
 - The NHS likely prioritized **infectious disease preparedness beyond the immediate pandemic response.**
 - Reduced funding for **Musculo Skeletal System Problems suggests shifting focus in healthcare delivery models.**
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5. Next Steps

- **Further Drilling Down** into specific categories for deeper insights.
- **Forecasting NHS expenditure trends** to predict future allocation priorities.
- **Comparative analysis** with international healthcare spending models for benchmarking NHS efficiency.

This structured approach ensures that the NHS expenditure analysis is **comprehensive, data-driven, and effectively communicated to stakeholders.**